Multimodal Skeleton-Based Sign Language Recognition using Spatial-Temporal Graph Convolutional Networks

Navadheer Vulli, Akshay M, Kesava Korlapati
Indian Institute of Technology Mandi
Emails: b23361@iitmandi.ac.in, b23337@iitmandi.ac.in, b23349@iitmandi.ac.in

***Abstract*—Sign language recognition (SLR) is a challenging problem in computer vision due to the complex articulation of human body joints and the temporal dynamics of gestures. In this work, we propose a multimodal skeleton-based approach using Spatial-Temporal Graph Convolutional Networks (ST-GCN) to effectively model sign language sequences. Instead of relying on raw RGB inputs, we represent human pose as a structured graph where joints correspond to nodes and anatomical connections correspond to edges.**

To capture complementary aspects of motion and structure, we design a multimodal framework consisting of four feature streams: joint positions, bone vectors, joint motion, and bone motion. Each stream captures distinct information about spatial configuration and temporal evolution. These streams are processed independently using graph convolutional networks and their predictions are fused using a weighted ensemble strategy.

Experiments on the large-scale WLASL2000 dataset demonstrate that the proposed approach significantly improves recognition performance. The ensemble model achieves approximately 49% Top-1 accuracy and 81% Top-5 accuracy, highlighting the effectiveness of multimodal fusion and graph-based representation for sign language recognition.

I. Introduction

Sign language serves as a primary communication medium for the hearing-impaired community. Automatic sign language recognition aims to translate gestures into textual or spoken language, thereby enabling seamless interaction.

Traditional approaches rely on RGB-based convolutional neural networks (CNNs). While these methods achieve reasonable performance, they suffer from several limitations:

- Sensitivity to lighting and background variations
- High computational cost
- Difficulty in capturing fine-grained motion

To overcome these challenges, skeleton-based approaches have gained popularity. By focusing on human pose keypoints, these methods provide a compact and robust representation of motion.

However, a single skeletal representation cannot fully capture the complexity of sign language. For example:

- Joint positions capture spatial structure but ignore motion dynamics
- Motion features capture dynamics but ignore structural relationships

Therefore, we propose a **multimodal approach** that integrates multiple skeletal representations to capture diverse aspects of sign language.

Key Contributions:

- A multimodal skeleton-based representation using four feature streams
- Graph-based modeling of spatial and temporal dependencies
- A weighted ensemble framework for improved recognition accuracy

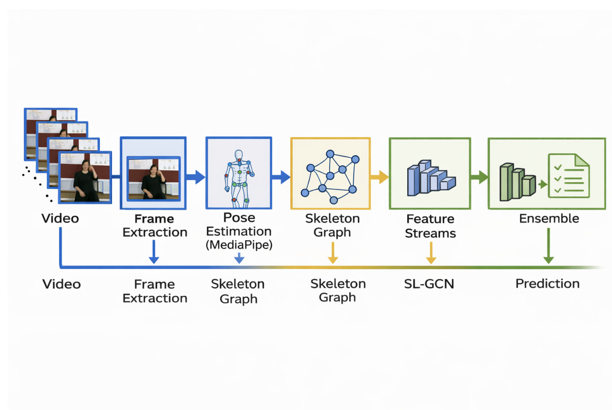

Fig. 1. Overall Pipeline

II. Dataset

We evaluate our approach on the WLASL2000 dataset, one of the largest datasets for word-level sign language recognition.

- Number of classes: ∼2000 signs
- Number of samples: ∼21,000 videos
- Number of signers: 100+

The dataset includes variations in:

- Signing speed
- Camera viewpoint
- Background conditions

These factors make WLASL2000 a challenging benchmark for recognition systems.We use the WLASL dataset [2].

III. Data Preprocessing

Each video is processed through the following pipeline:

1) Frame extraction from video
2) Pose estimation using MediaPipe
3) Skeleton graph construction
4) Data normalization and formatting

Each sample is represented as:

$$X \in \mathbb{R}^{C \times T \times V \times M}$$

where:

- C = number of channels (x, y, z, confidence)
- T = number of frames
- V = number of joints (27)
- M = number of persons (1)

Skeletons are extracted using pose estimation methods [6].

IV. Methodology

A. Multimodal Feature Design

To capture diverse aspects of sign language, we construct four modalities.

1) Joint Representation:

$$v_{i,t}^{J} = (x_{i,t}, y_{i,t}, z_{i,t}, s_{i,t})$$

This captures spatial positions of joints.

2) Bone Representation:

$$v_{i,t}^{B} = (x_{j,t} - x_{i,t}, y_{j,t} - y_{i,t}, z_{j,t} - z_{i,t}, s_{j,t})$$

This captures structural relationships.

3) Joint Motion:

$$v_{i,t}^{JM} = v_{i,t+1}^{J} - v_{i,t}^{J}$$

This models temporal movement.

4) Bone Motion:

$$v_{i,t}^{BM} = v_{i,t+1}^{B} - v_{i,t}^{B}$$

This captures structural dynamics.

Intuition:

- Joint $\rightarrow$ spatial structure
- Bone $\rightarrow$ relational structure
- Motion $\rightarrow$ temporal dynamics

Multi-stream representation improves performance as shown in [4].

B. Graph Construction

The skeleton is modeled as a graph:

- Nodes: body joints
- Edges: natural anatomical connections

Adjacency matrix:

$$A \in \mathbb{R}^{V \times V}$$

V. Model Architecture

A. Graph Convolution

$$X_{out} = D^{-1/2} A D^{-1/2} X W$$

Explanation:

- A defines connectivity
- D normalizes the graph
- W learns feature transformations

This allows information sharing between neighboring joints.Graph convolution is based on ST-GCN [3].

B. Temporal Modeling

Temporal modeling captures the dynamic evolution of gestures across video frames. Temporal convolution is applied along the time dimension to learn motion patterns such as hand trajectories and transitions. It enables efficient extraction of both short-term and long-term dependencies without sequential processing. In skeleton-based representations, it models the movement of joints over time, helping distinguish similar static poses with different motions. This improves the recognition of fine-grained sign language gestures.

C. Overall Architecture

- Each modality is processed independently
- Outputs are fused using ensemble

The proposed system follows a **multi-modal and multi-stream architecture**, where different representations of the same sign language video are treated as independent sources of information. Each modality captures a unique aspect of the signing process, such as spatial structure, motion dynamics, or appearance features.

First, the input video is converted into multiple modalities, primarily focusing on **skeleton-based representations**. Using a pose estimation model, human keypoints are extracted from each frame, forming a structured representation of the human body. These keypoints are then transformed into four distinct streams:

- **Joint Stream:** Represents the absolute positions of keypoints $(x, y, [z], s)$, capturing spatial structure.
- **Bone Stream:** Represents relative vectors between connected joints, encoding structural relationships of the human body.
- **Joint Motion Stream:** Captures temporal changes in joint positions across frames, modeling motion dynamics.
- **Bone Motion Stream:** Captures temporal variations in bone vectors, highlighting dynamic structural changes.

Each of these streams is processed independently using a dedicated **Sign Language Graph Convolutional Network (SL-GCN)**, which models both spatial relationships (via graph connections between joints) and temporal dependencies (across video frames). This independent processing allows each stream to specialize in learning specific patterns without interference. The key motivation behind this design is that

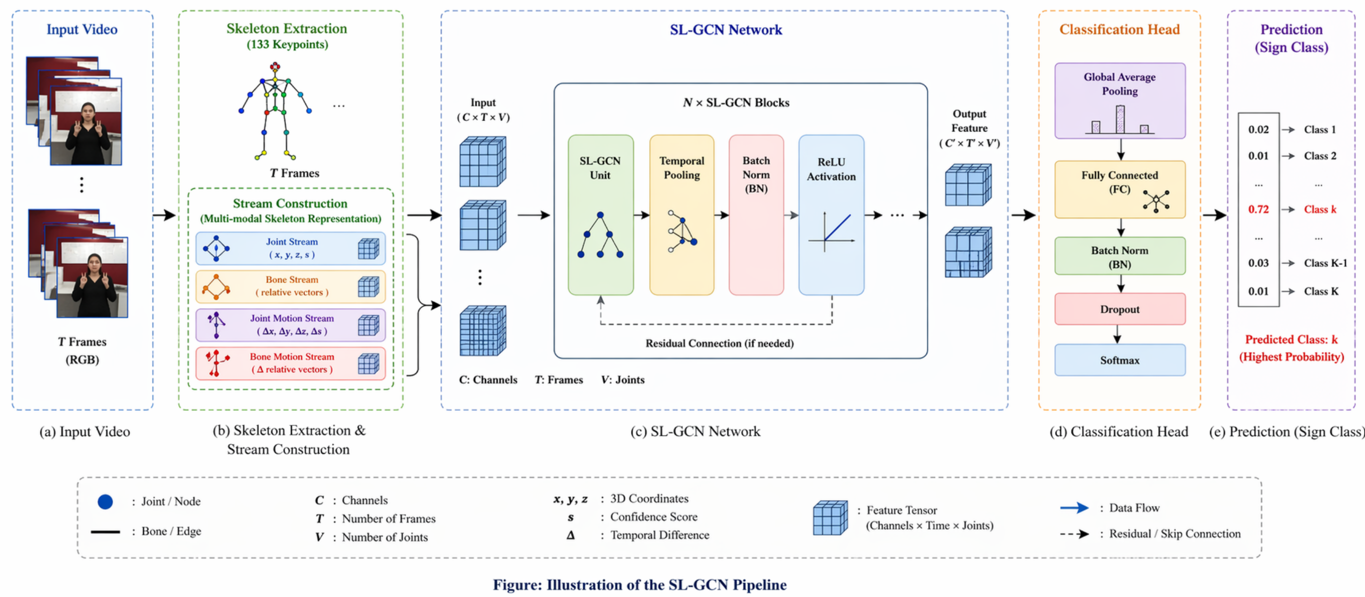

Fig. 2. Illustration of SL-GCN multi-stream pipeline including joint, bone, motion streams and attention modules.

different modalities capture complementary information. For example:

- Joint features capture absolute pose
- Bone features capture relative geometry
- Motion streams capture temporal dynamics

After feature extraction and classification from each stream, the predictions are combined using a **late fusion ensemble strategy**. Instead of merging raw features, the system combines the **prediction scores** from each stream. This ensures that each modality contributes to the final decision based on its learned confidence.

Mathematically, the final prediction is obtained as a weighted combination of all streams.

This ensemble strategy improves robustness and accuracy, as errors from one modality can be compensated by others. It also aligns with multi-modal learning principles, where combining diverse representations leads to better generalization.

Overall, the architecture effectively integrates spatial, structural, and temporal information, making it highly suitable for complex tasks like sign language recognition where fine-grained motion and body coordination are critical. :contentReference[oaicite:0]index=0

All streams are trained independently using:

- Loss: Cross-entropy
- Optimizer: SGD with momentum
- Batch size: 16
- Epochs: 100

$$\mathcal{L} = -\sum y \log(\hat{y})$$

VI. Ensemble Strategy

Final prediction is computed as:

$$Score = \frac{\sum_{i=1}^{4} \alpha_i S_i}{\sum \alpha_i}$$

where:

- s_i represents the prediction scores from each stream
- α_i represents the importance weight assigned to each modality

Why ensemble works:

- Reduces model variance
- Combines complementary features
- Improves robustness

Attention mechanisms enhance feature learning [5].

VII. Experimental Results

Model	Top-1 (%)	Top-5 (%)
Joint	42	74
Bone	27	55
Joint Motion	18	38
Bone Motion	19	40
Ensemble	**49**	**81**

TABLE I
Performance on WLASL2000

A. Discussion

- Joint stream performs best individually
- Motion streams capture temporal variations
- Ensemble improves overall performance

VIII. Challenges

- Similar gestures across classes
- Fine-grained finger movements
- Pose estimation inaccuracies

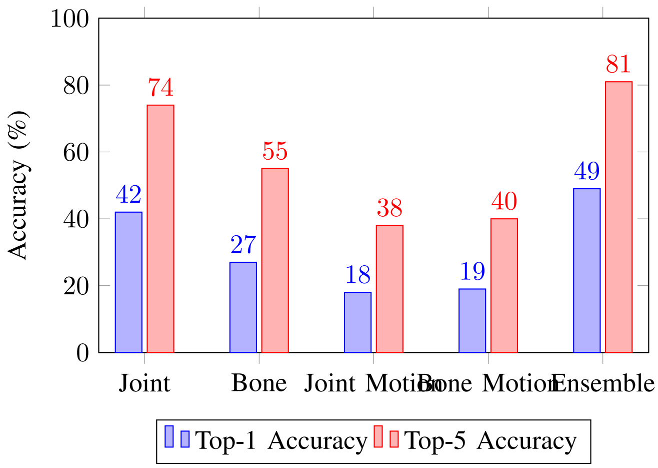

Fig. 3. Comparison of Top-1 and Top-5 accuracy across different streams.

IX. Limitations

- Moderate Top-1 accuracy
- Limited hand-level precision
- Dependency on skeleton quality

X. Conclusion

In this work, we presented a multimodal skeleton-based approach for sign language recognition using graph neural networks. By integrating multiple skeletal modalities and leveraging spatial-temporal modeling, the system effectively captures complex motion patterns. The ensemble strategy significantly improves performance, demonstrating the importance of multimodal learning in sign language recognition.

References

[1] D. Li, R. Wang, and H. Li, "Multimodal Sign Language Recognition with Skeleton-based Graph Neural Networks," *arXiv preprint arXiv:2107.xxxxx*, 2021.

[2] D. Li, C. Rodriguez, X. Yu, and H. Li, "Word-Level Deep Sign Language Recognition from Video," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 1–9.

[3] S. Yan, Y. Xiong, and D. Lin, "Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.

[4] L. Shi, Y. Zhang, J. Cheng, and H. Lu, "Two-Stream Adaptive Graph Convolutional Networks for Skeleton-Based Action Recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[5] C. Si, W. Chen, W. Wang, L. Wang, and T. Tan, "An Attention Enhanced Graph Convolutional LSTM Network for Skeleton-Based Action Recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[6] F. Zhang et al., "MediaPipe Hands: On-device Real-time Hand Tracking," *Google AI Blog*, 2020.

[7] Z. Cao, T. Simon, S.-E. Wei, and Y. Sheikh, "Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.